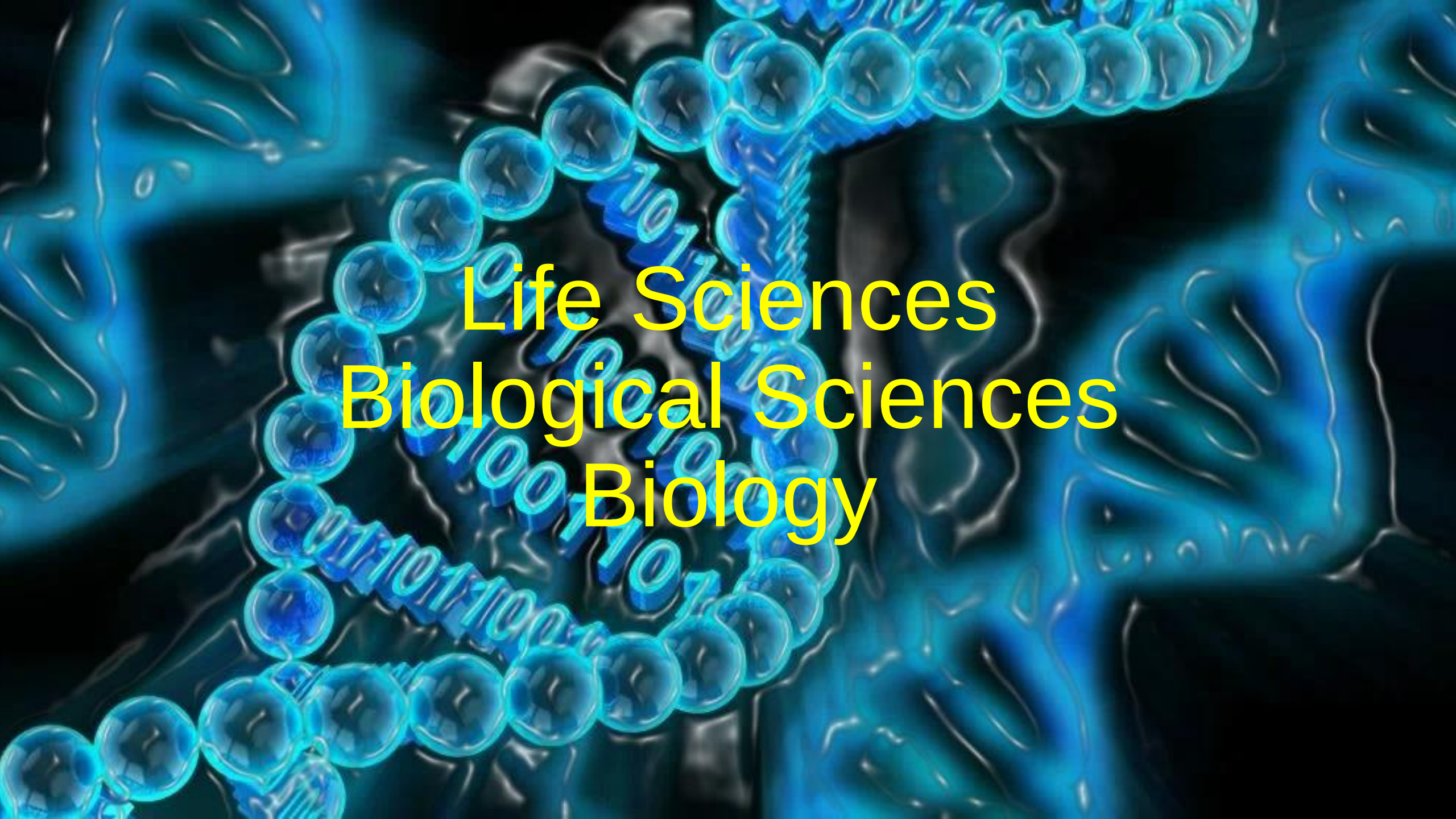




Life Sciences
Biological Sciences
Biology

About Myself

- Dr. Zhou YANG (杨洲), yangzhou@cuhk.edu.cn, Dr. Yang
- BSc. and Ph.D. in Hong Kong University of Science and Technology
- Assistant Professor (Teaching), joined CUHKSZ in June, 2020
- The Presidential Exemplary Teaching Award 2021
- General Biology, Chemistry and Life Sciences, Biochemistry Laboratory, Animal Physiology, Neurobiology
- Systems Bioengineering II: Neural Systems, Cell and Molecular Biology, Anatomy/Physiology

Why students from SDS should learn **Biology** , and, in **English**?

- Life sciences are among the **most data-rich**, societally important, and methodologically challenging domains.
- Provide the context, data types, and problem structures needed to build useful models, design rigorous studies, and make real-world impact.
- Bottom line: Health

- Data diversity and volume: Life sciences span sequences, images, time series, graphs, and spatial data at massive scale.
- Societal impact: Skills translate directly to healthcare, public health, biotech, agriculture, and climate—areas with clear human and economic benefit.
- Career opportunities: Biopharma, med-tech, digital health, diagnostics, and research institutes actively hire data scientists for these problems (genomics, imaging, wearables, clinical decision support).
- Ethics and governance: Human data introduces privacy, consent, bias, and equity considerations—core to responsible AI.
- Interpretability and mechanism: Biological knowledge (pathways, cell types, anatomy) helps build interpretable models and knowledge-guided AI, a growing industry need.
- Collaboration skills: Working with biologists, clinicians, and lab teams improves communication and problem framing.

Questions raised by your predecessors

- Cannot see the white board
 - Use only slides
- Too difficult for students who have not learned biology before
 - Survey, content adjust accordingly
- Too much contents
 - Streamlining course content
- Speak too fast
 - Slow Down, welcome to interrupt

  Information

▼ BIO1008:Chemistry and Life Sciences_L01L02L03 

Home Page

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Content

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COURSE MANAGEMENT

▼ Control Panel

Information 

Build Content ▼ Assessments ▼ Tools ▼ Partner Content ▼



Course Schedule Dates

Enabled: Statistics Tracking

Select your class date

Biology Background 

This survey aims to understand the students' background in Biology so that the course instructor can tailor the course content accordingly.

Questions and Office Hours

- Just **Raise Your Hand (if too fast)**
- Q&A session, 5 - 10mins each class
- Office hours
 - ❖ Mon & Wed 09:00-10:00 TD525
 - ❖ by email appointment



Course Format

- Lectures
- 2 Assignments
- Final Exam (excluding the content covered in the first part)

Grading

	Percentage
Assignments (x2)	10
Final Exam	40

Textbook

Essential Biology with Physiology (6th Global edition, 2020)

**By: Eric J. Simon, Jean L. Dickey,
Kelly A Hogan, and Jane B. Reece**

Course Content Mapping

❖ Cells and Genetics

- ✓ Biomolecules: *structure prediction (AlphaFold)*
- ✓ Cell structures: *Quantitative microscopy*
- ✓ The Working Cell: *Enzyme kinetics*
- ✓ Cellular respiration: *metabolic flux analysis*
- ✓ Cell Reproduction and Cancer: *TCGA*
- ✓ The structure and function of DNA: *viral epidemiology*
- ✓ Gene expression: *regulatory genomics*
- ✓ DNA Technology: *Sequencing*

❖ Animal Structure and Function

- ✓ Nervous, sensory, and locomotor systems, *EEG*
- ✓ Circulation and respiration: *Imaging analytics: CT/MRI*
- ✓ Immune system, *vaccine response modeling*
- ✓ Hormones: *endocrine modeling*
- ✓ Nutrition and digestion, *Microbiome analytics*

Learning Outcomes

- Explain what it means to say that an organism is "alive."
- Explain what organisms are made of and how cells come to be.
- Explain how organisms process hereditary information and acquire and use energy.
- Explain the relationships between species and how evolution occurs.
- Explain the scientific process.

What Does It Mean to Say That Something is Alive?

All living organisms share five fundamental characteristics:

1. **Cells:** All organisms are made up of membrane-bound cells
2. **Replication:** ^{复制} All organisms are capable of reproduction
3. **Information:** All organisms process hereditary information encoded in **genes** as well as information from environment
4. **Energy:** All organisms acquire and use energy to stay alive
5. **Evolution:** Populations of organisms are continually evolving

Theories

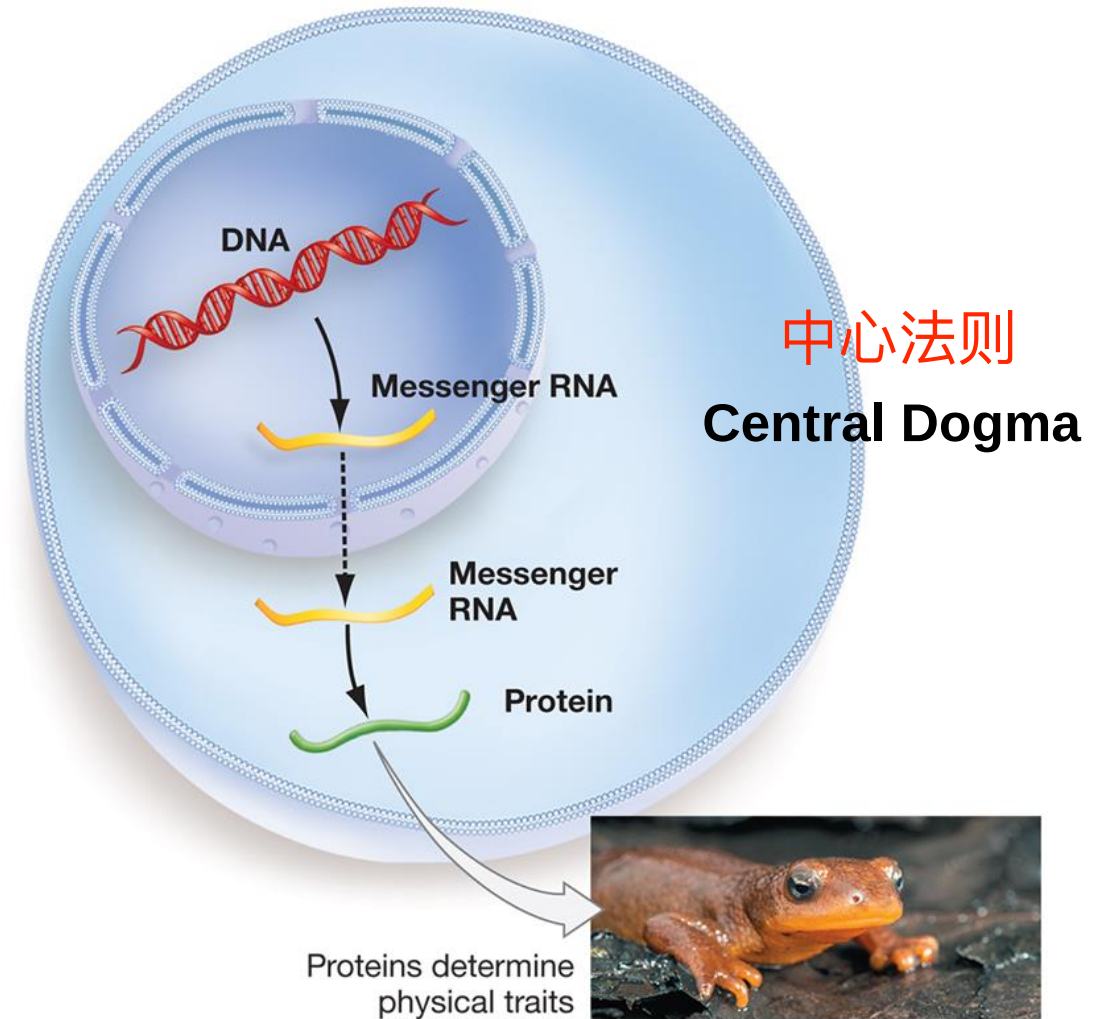
- **Theory:**
 - Explanation for a very general class of phenomena or observations supported by wide body of evidence:
- **Three theories** form framework for modern biological science:
 - **Cell theory:**
 - What are organisms made of? Where do organisms come from?
 - **Chromosome theory of inheritance:** 染色体遗传学说
 - How is hereditary information transmitted from one generation to the next?
 - **Theory of evolution by natural selection:**
 - How are organisms related to one another?

Cell theory

- All organisms are made up of cells
 - 1660s—Robert Hooke and Anton van Leeuwenhoek were first to observe **cells**
 - Highly organized compartments
 - Separated from their environment by membrane barrier 膜屏障
- All cells come from **preexisting cells** (must replicate)
 - As predicted by **cell theory**: 既有细胞（或已存在的细胞）
 - All cells present in multicellular organism have descended from preexisting cells
- Evidence now shows that life arose from non-life early in Earth's history by process of **chemical evolution**

Chromosomal theory of inheritance

- Proposed by Sutton and Boveri:
 - Hereditary ^{遗传的 / 先天的} or genetic information is encoded in genes
 - **Genes** are units located on chromosomes
- 1950s—**Chromosomes** are molecules of **deoxyribonucleic acid (DNA)**:
 - DNA is the hereditary material
 - Genes are segments of DNA that code for cell products



Theory of evolution by natural selection

- Darwin and Wallace made two claims regarding natural world:
 - Species are related by **common ancestry**: 共同祖先 (或共同祖源)
 - Characteristics of species can be modified from generation to generation:
 - Darwin called this “**descent with modification**”
改良的后代 / 逐渐演化的后裔 (常译作 “有改造的遗传”)
- **Evolution:**
 - Change in characteristics of population over time
 - Species are related to one another and can change through time
 - **Population:**
 - Group of individuals of same species
 - Living in the same area at the same time

Natural selection explains how evolution occurs

- **Two conditions** must be met for natural selection to occur in a population:
 1. Individuals must vary in characteristics that are **heritable**—can be passed onto offspring
可遗传的 / 可遗传性
 2. In particular environment, certain versions of these heritable traits help individuals reproduce more than other versions
- Certain **heritable traits** lead to increased success in producing offspring:
 - Traits become more common in population over time
- **Evolutionary change** occurs in **populations**
- **Speciation** occurs when populations **diverge** to form new species
分歧 / 演化分化

Fitness and Adaptation Drive Natural Selection

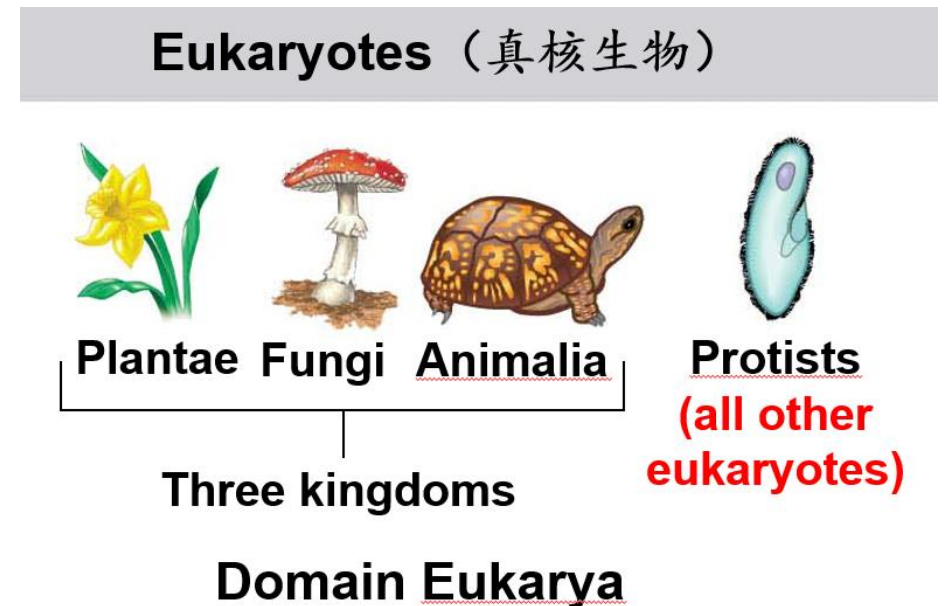
- **Natural selection** acts on **individuals**
- **Fitness:**
 - Ability of individual to produce offspring:
 - Individuals with high fitness produce many more surviving offspring than do others in population
- **Adaptation:**
 - Trait that **increases fitness** of individual in particular environment

DS: Analyzing Genetic Variation

- How is **genetic variation** analyzed?
- Biologists study RNA and DNA from different organisms:
 - Compare sequences of building blocks (A,T,C,G)
 - **Fewer** sequence variations between two species may indicate **closer** relationship
- Example:
 - Land plant DNA: A – T – A – T – C – G – A – G
 - Green algae DNA: A – T – A – T – G – G – A – G
 - Brown algae DNA: A – A – A – T – G – G – A – C
 - **Green algae** is more closely related to land plants than brown algae

Life in Its Diverse Forms

- Biologists add thousands of newly identified species to the list each year.
- Estimates of the total number of species (种系) range from 10 million to more than 100 million.
- A **species** is generally defined as a group of organisms that have the potential to **interbreed** with one another in nature to produce healthy offspring.
交配 / 相互交配
- The **three domains** of life are
 - Bacteria (细菌), almost everywhere
 - Archaea (古细菌), many in extreme environments
 - Eukarya (真核生物界)



Doing Biology: scientific process

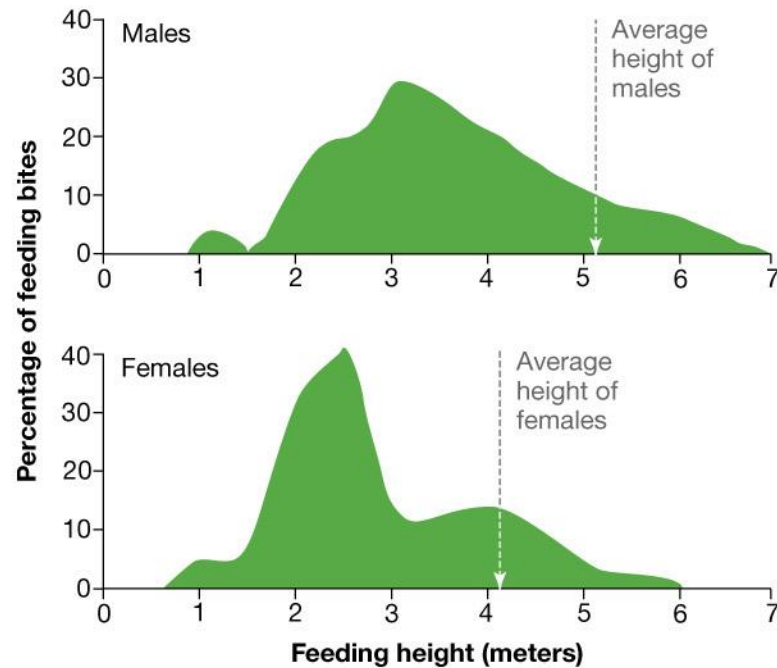
- The nature of science:
 - All scientists ask questions that can be answered by measuring things—by collecting data
- Science involves:
 - **Formulating hypotheses**
 - **Finding evidence** that supports or conflicts with those hypotheses
- **Hypothesis testing**—two-step process:
 1. State hypothesis as precisely as possible and list **predictions** it makes
 2. Design observational or experimental study capable of **testing** those predictions

Why Do Giraffes Have Long Necks?

- **Food competition hypothesis:**
 - **Argues** that long necks evolved because those with long necks can reach food unavailable to other mammals
 - **Predictions:**
 - Neck length variable among giraffes
 - Neck length in giraffes heritable
 - Giraffes feed high in trees

Giraffes Do Not Usually Extend Their Necks Upward to Feed

(a) Most feeding is done at about shoulder height.



(b) Typical feeding posture in giraffes



The Sexual Competition Hypothesis

- **Alternative hypothesis**—giraffes evolved long necks because:
 - Longer-necked males win more fights than shorter-necked males
 - Longer-necked males can then father more offspring
- Data support this hypothesis
- Data refutes food competition hypothesis
- Question is not closed
- All hypotheses must be tested rigorously



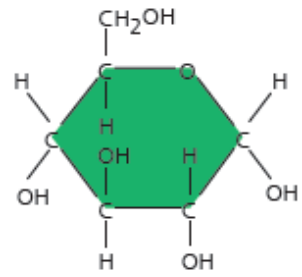
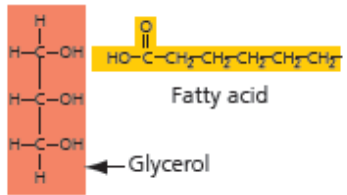
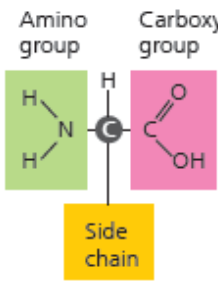
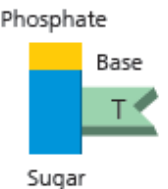
Terminology

- **Theory** refers to explanations for broad patterns
- **Hypothesis:**
 - Testable statement that explains something observed
- **Experiment**—allows researchers to test effect of a single, well-defined factor on particular phenomenon
- **Prediction**—measurable or observable result
 - Must be correct if hypothesis is valid

The Molecules of Life

Unit 1; Chapter 3

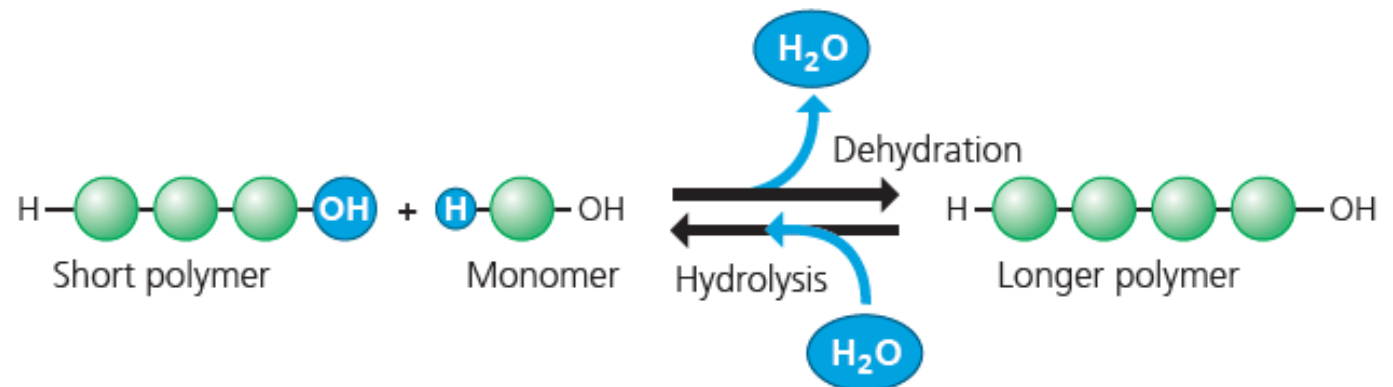


Large Biological Molecules	Functions	Components	Examples
Carbohydrates	Dietary energy; storage; plant structure	 Monosaccharide	Monosaccharides: glucose, fructose; disaccharides: lactose, sucrose; polysaccharides: starch, cellulose
Lipids	Long-term energy storage (fats); hormones (steroids)	 Fatty acid Glycerol Components of a triglyceride	Fats (triglycerides); steroids (testosterone, estrogen)
Proteins	Enzymes, structure, storage, contraction, transport, etc.	 Amino group Carboxyl group Side chain Amino acid	Lactase (an enzyme); hemoglobin (a transport protein)
Nucleic acids	Information storage	 Phosphate Base Sugar Nucleotide	DNA, RNA

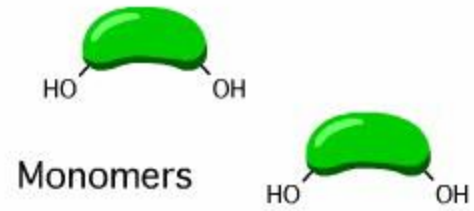
Giant Molecules from Smaller Building Blocks

- LO: Compare a **dehydration** (脱水) reaction to a **hydrolysis** (水解) reaction.
- **Macromolecules** are **polymers** (聚合物).
 - Carbohydrates, Proteins, Nucleic acids
- Polymers are made by stringing together many smaller molecules called **monomers** (单体).

Giant Molecules from Smaller Building Blocks



Animation: Polymers



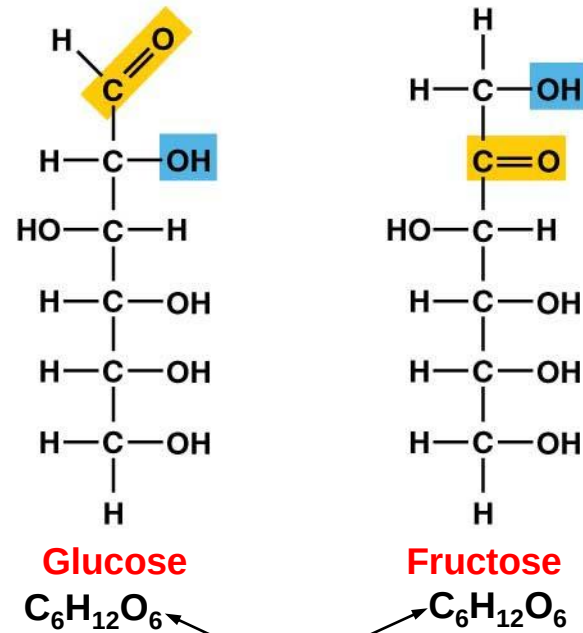
Carbohydrates (碳水化合物)

- LO: Compare the structures and roles of monosaccharides, disaccharides, and polysaccharides in living organisms. Give examples of each.
- Include sugars and polymers of sugar:
 - small sugar molecules in soft drinks
 - long starch molecules in spaghetti and bread
- **Mono**saccharides, **Di**saccharides, **Poly**saccharides
- In animals
 - a primary source of dietary energy
 - raw material for manufacturing other kinds of organic compounds
- In plants, as a building material for much of the plant body

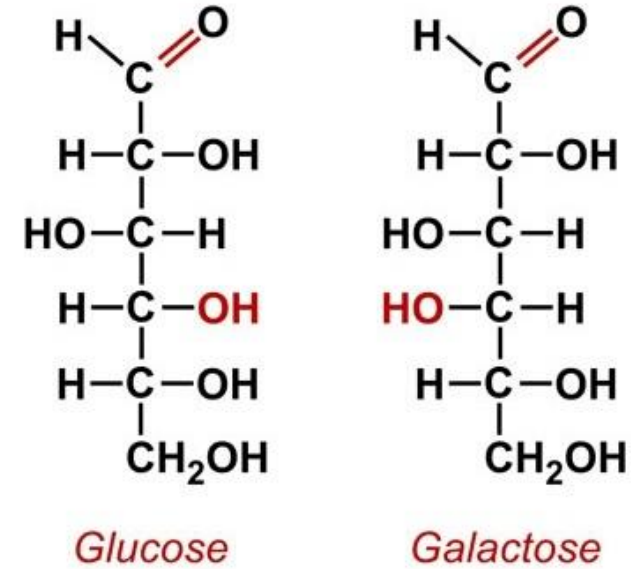
Monosaccharides (单糖)

- Monomers of carbohydrates
- Cannot be broken down into smaller sugars
- General formula $(CH_2O)_n$, where n is usually 3, 4, 5, 6, 7, or 8
- **Glucose** (葡萄糖) in sports drinks
- **Fructose** (果糖) found in fruit.
- Glucose and fructose are **isomers** (同分异构体), molecules that have the same molecular formula but different structures.

Fructose tastes about 1.7 times sweeter than glucose.



Isomers
 (same formula, different arrangements)



Galactose is only about half as sweet-tasting as glucose.

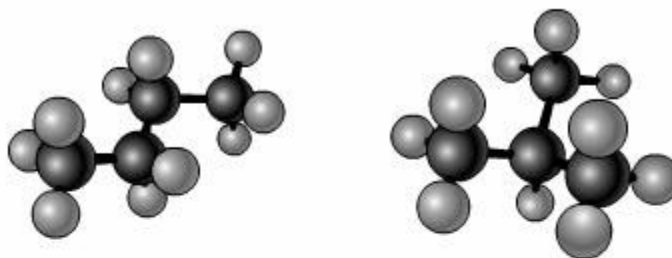


Celery

Animation: Isomers

结构异构

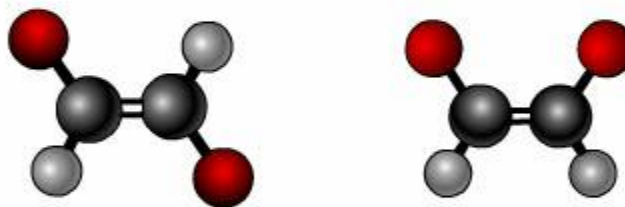
Structural isomers



几何异构

Geometric isomers

(Cis-trans isomers)



对映异构

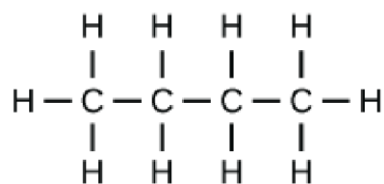
Enantiomers

Stereoisomers

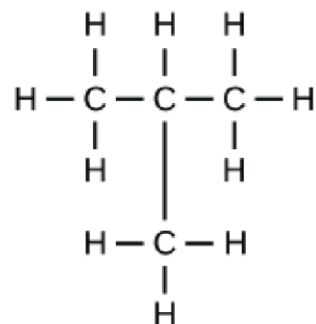


Structural isomers

Butane

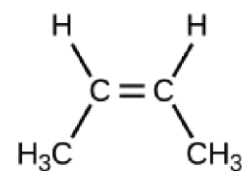


Isobutane



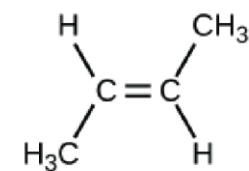
Geometric isomers

cis-2-butene



methyl groups on
same side of double bond

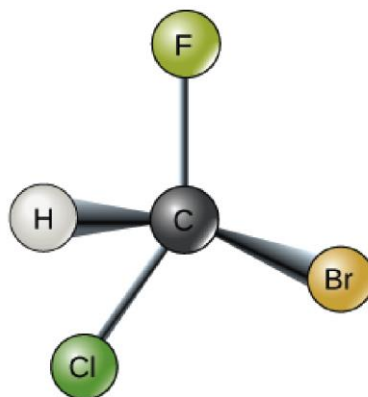
trans-2-butene



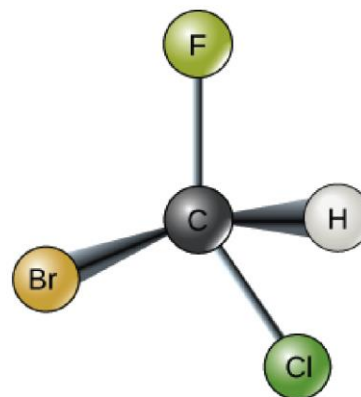
methyl groups on opposite
sides of double bond

Enantiomers

L-isomer

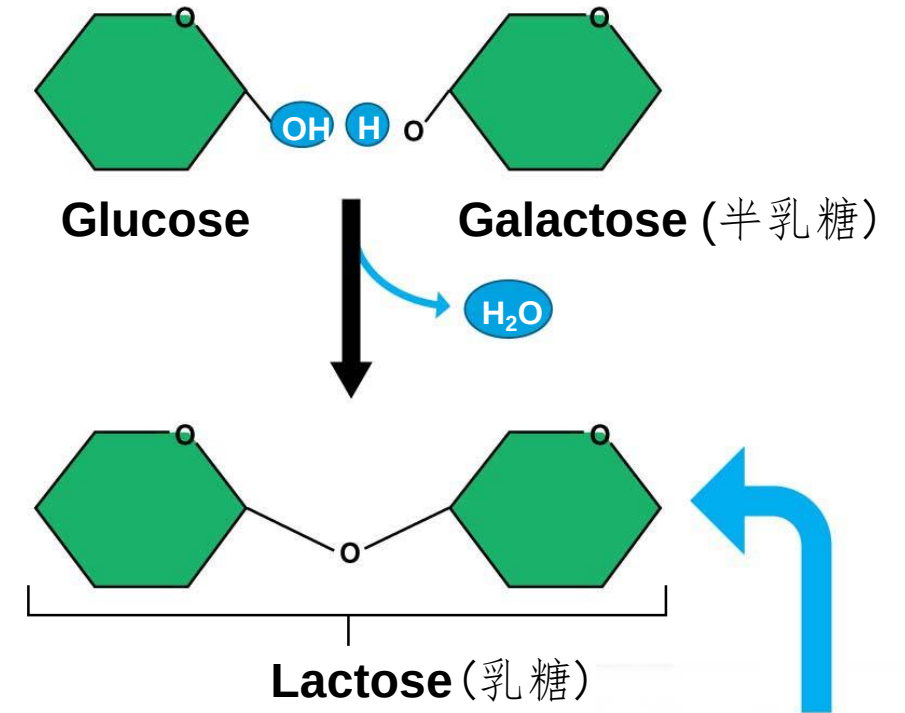


D-isomer



Disaccharides (双糖)

- Double sugar constructed from two monosaccharides by a dehydration reaction
- Lactose (乳糖) in milk, breast milk
- Maltose (麦芽糖) in beer
- Sucrose (蔗糖) in table sugar
 - A major cause of tooth decay
 - Overconsumption increases the risk of developing type 2 diabetes (糖尿病) and heart disease

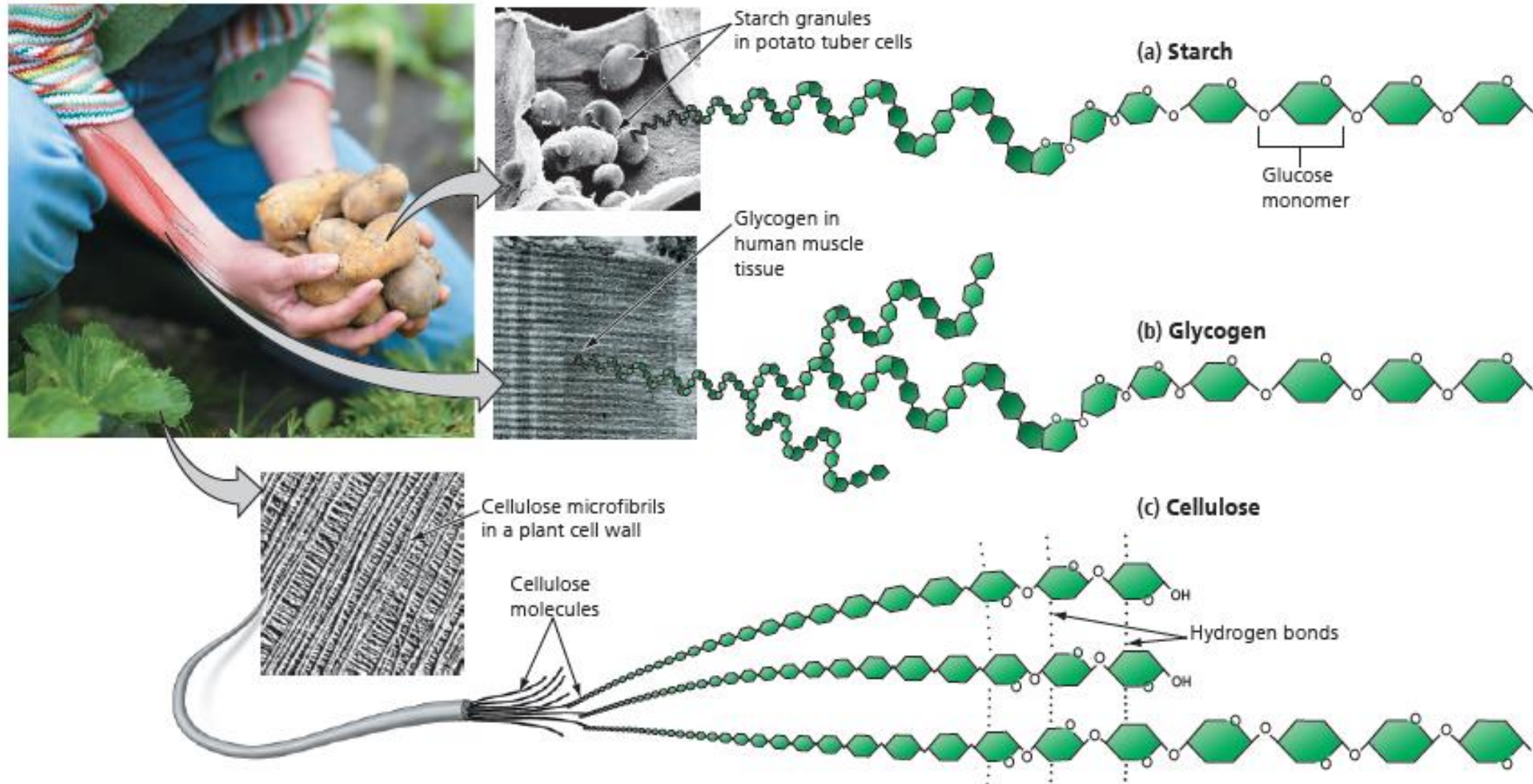




Lactose Intolerance (乳糖不耐受)

Polysaccharides (多糖) are Complex carbohydrates made of long chains of sugars—polymers of monosaccharides.

▼ **Figure 3.9** Three common polysaccharides.

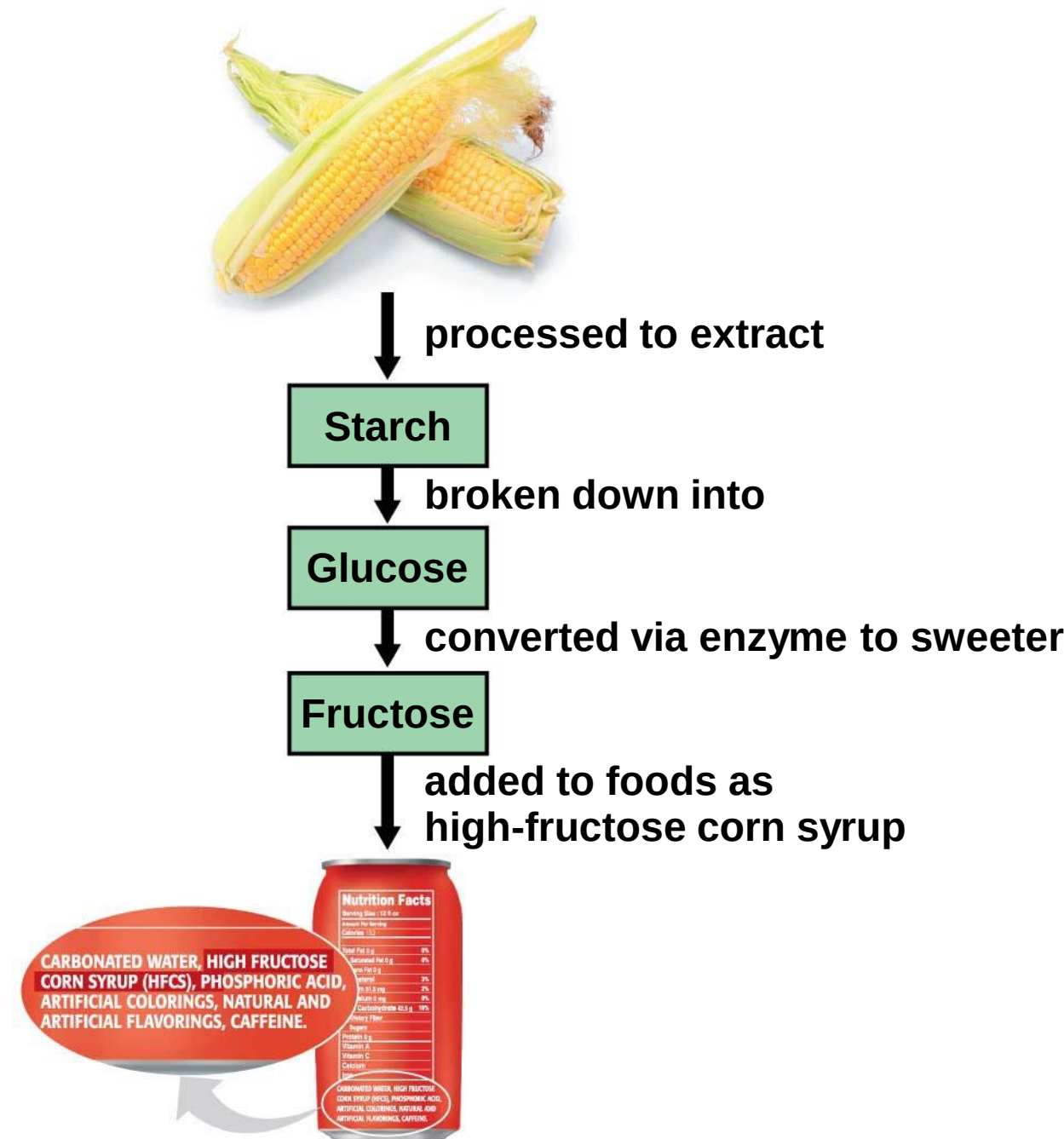


Polysaccharides (多糖)

- Starch (淀粉)
 - Used by plant cells to store energy
 - Potatoes and grains in our diet
- Glycogen (糖原)
 - Used by animal cells to store energy in muscle and **liver**
 - broken down to release glucose when you need energy
- Cellulose (纤维素)
 - Forms cable-like fibrils (原纤维) in the walls that enclose plant cells
 - The **most abundant organic compound** on Earth
 - Cannot be broken by any enzyme produced by animals

High-fructose corn syrup (HFCS) (高果葡糖浆)

- Common sweetener
- HFCS is often one of the first ingredients listed in soft drinks.
- 55% fructose
- To replace sucrose
- Why not sucrose?



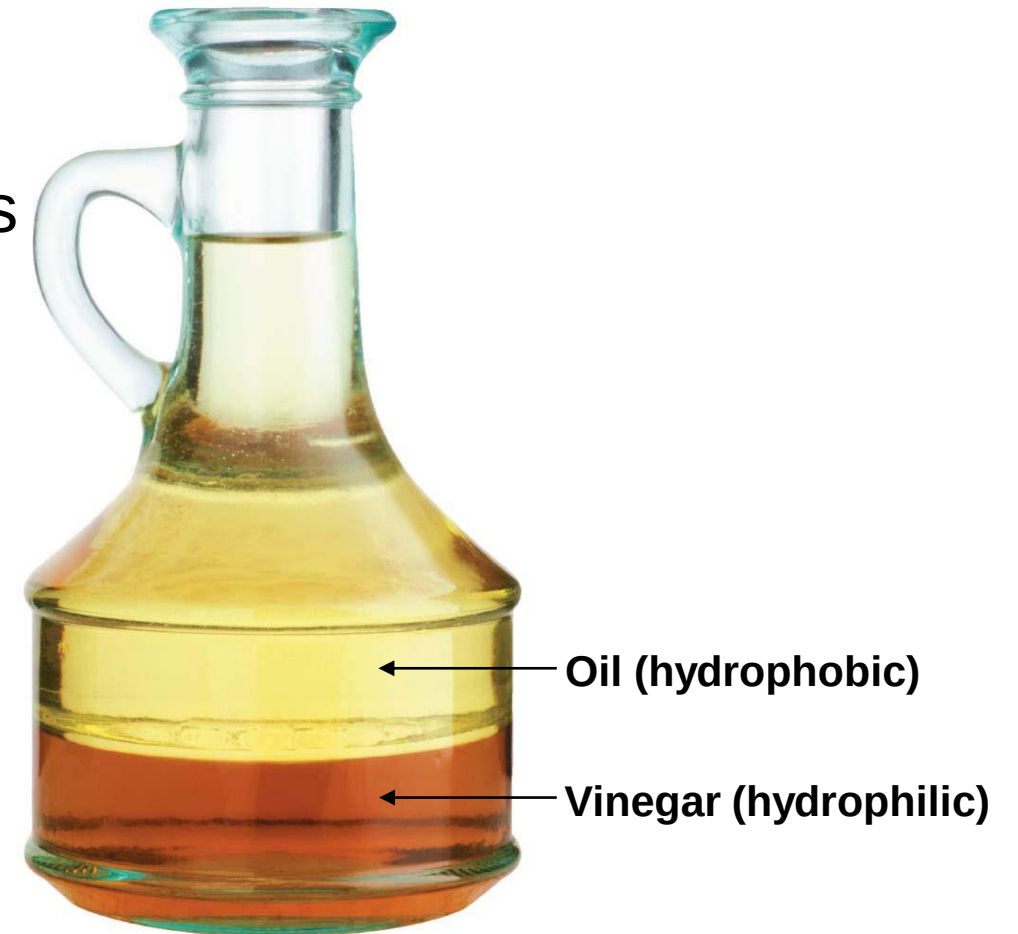
Lipids (脂类)

LO:

- Describe the unique properties of lipids.
- Compare the structure and properties of saturated and unsaturated fatty acids.
- Distinguish between steroids and anabolic steroids and explain how the use of anabolic steroids can be dangerous to a person's health.

Hydrophilic and Hydrophobic

- Hydrophilic (“water-loving”)
 - Philip likes to drink water
 - Dissolve readily in water
 - Carbohydrates, amino acids, nucleotides
- Hydrophobic (“water-hating”)
 - Lipids
 - Unable to mix with water
 - “Hydrophobic force”?

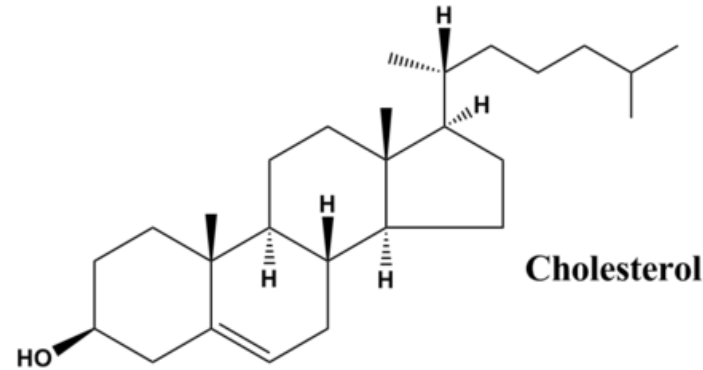


Lipids

- Lipids are a **diverse** group of molecules made from different molecular building blocks.
- Large (not giant) biological molecules.
- Not macromolecules (polymers built from repeating monomers)
 - Carbohydrates, by monosaccharides
 - Proteins, by amino acids
 - Nucleic acids, by nucleotides

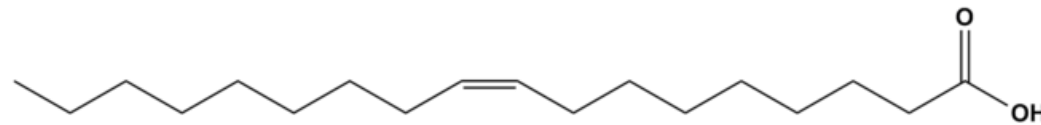
Diverse Lipids

- Fats



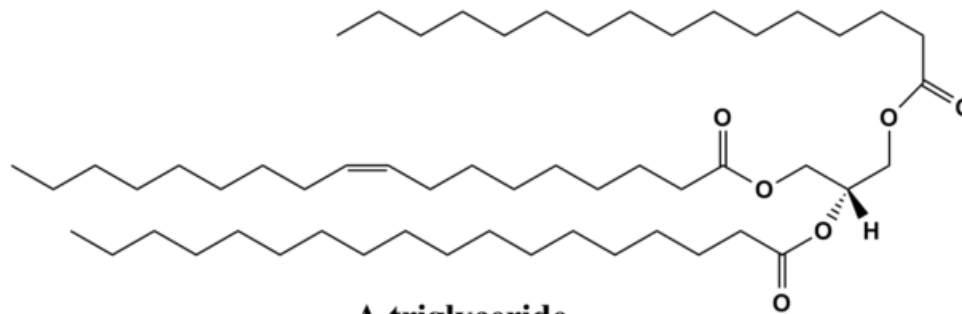
Cholesterol

- Steroids

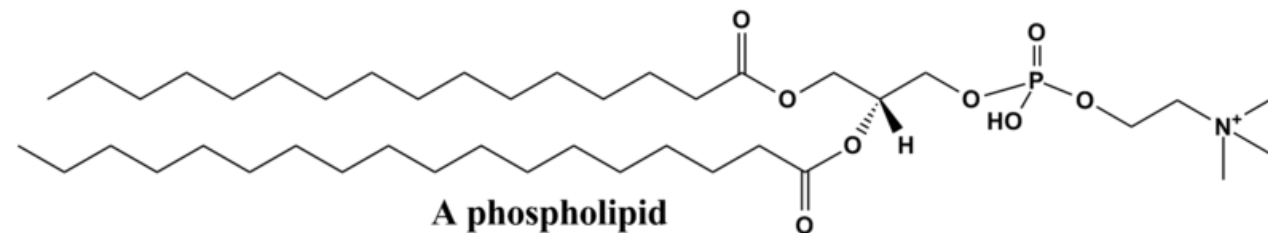


A free fatty acid

- Phospholipids



A triglyceride



A phospholipid

Fats

- or **triglyceride** (甘油三酯)
- Consists of a **glycerol** (甘油) molecule joined with **three fatty acid** (脂肪酸) molecules via a dehydration reaction
- Essential functions in the human body
 - Energy storage
 - Cushioning
 - Insulation (隔热, 保温)

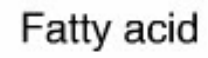
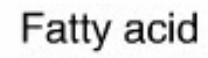
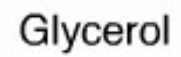
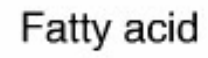
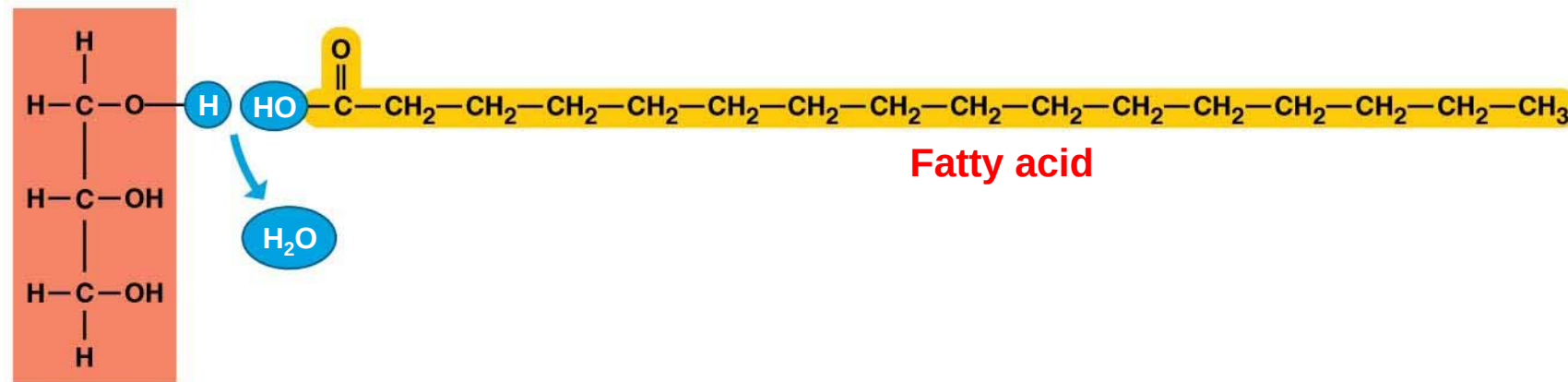
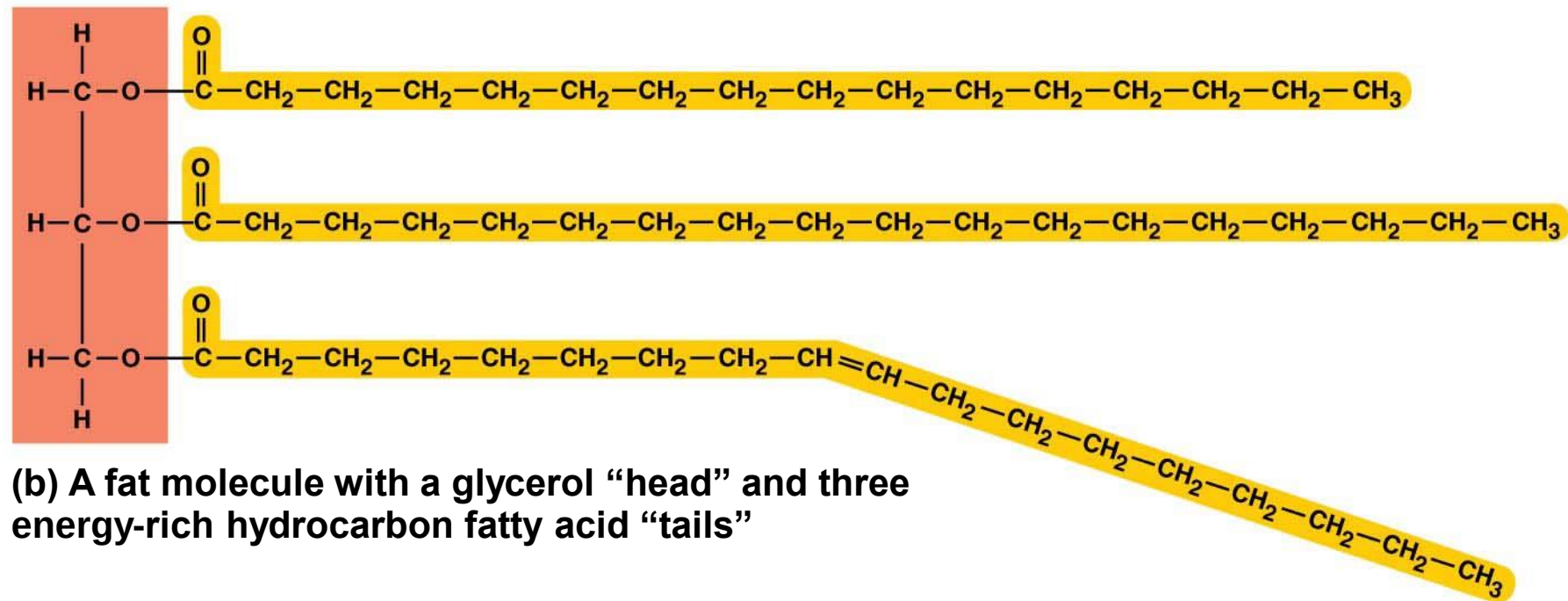


Figure 3.11



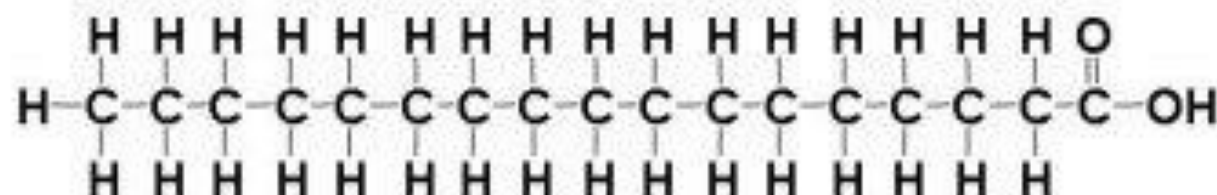
Glycerol (甘油)

(a) A dehydration reaction linking a fatty acid to glycerol

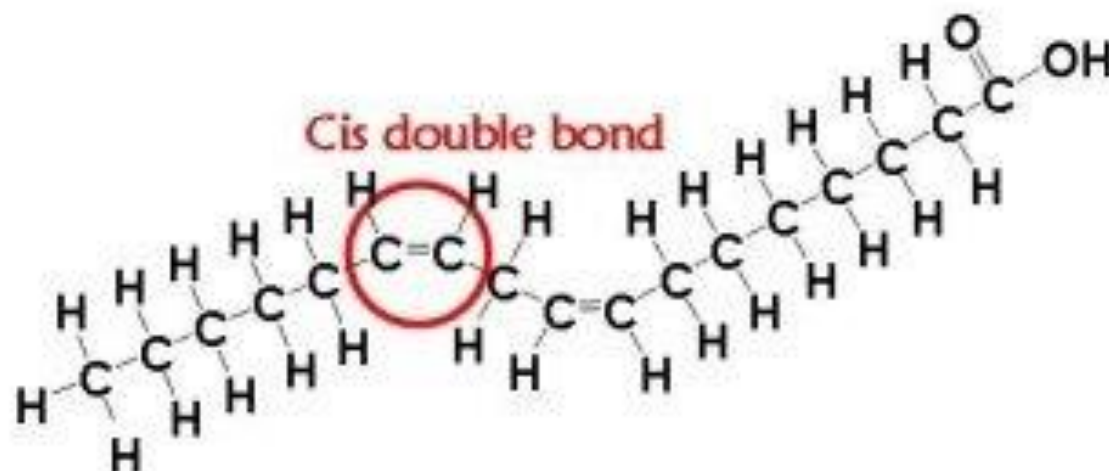


(b) A fat molecule with a glycerol “head” and three energy-rich hydrocarbon fatty acid “tails”

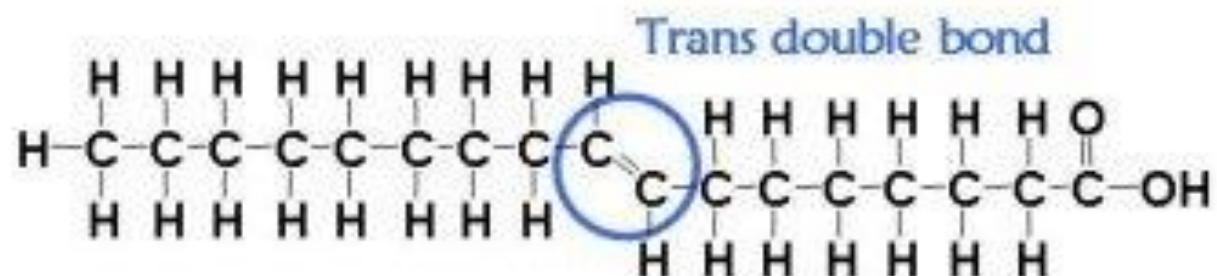
SATURATED
Stearic acid
(found in butter)



UNSATURATED
Linoleic acid
(found in vegetable oil)

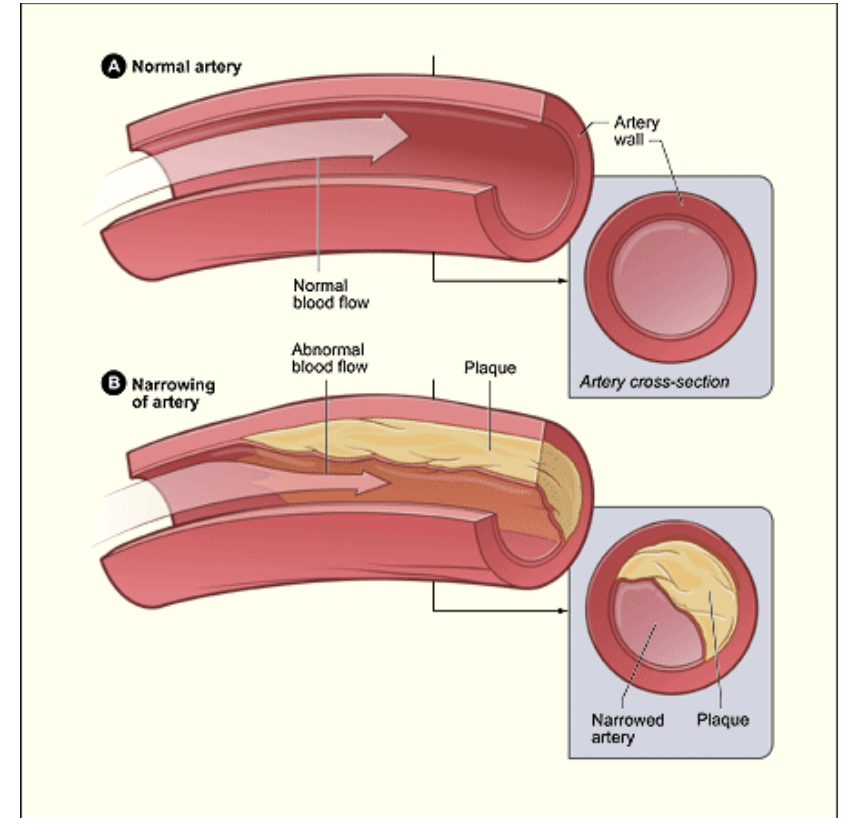


TRANS
trans-Linoleic acid
(found in some
margarine)



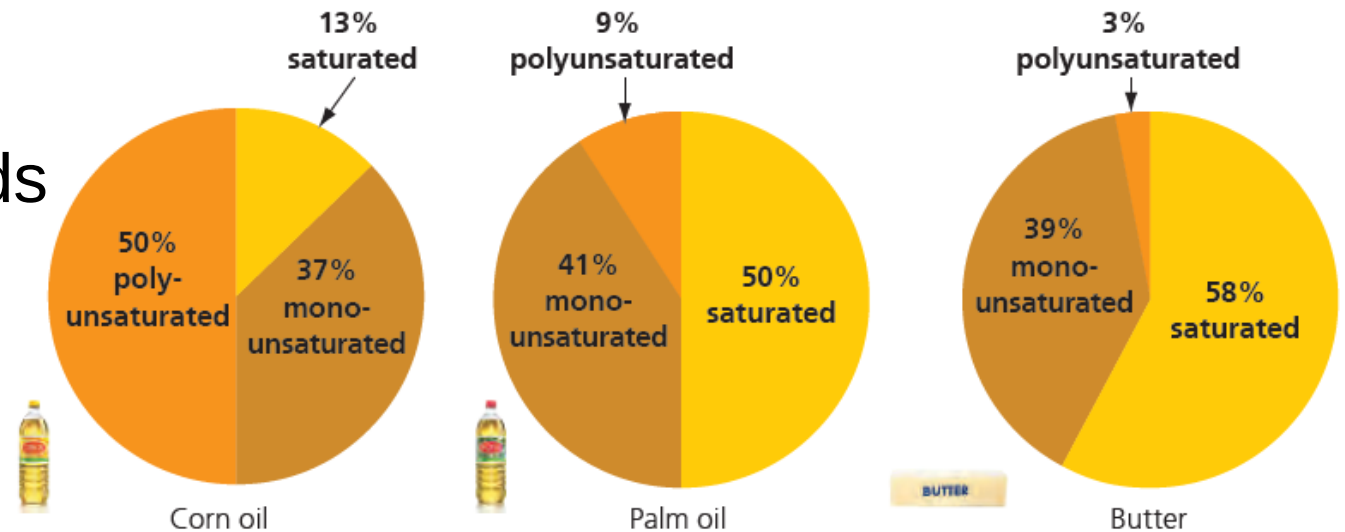
Saturated (饱和的) Fats

- **Saturated**, if it has the **maximum** number of hydrogens
 - A saturated fat has **all three** of its fatty acids saturated
- Most animal fats
 - A relatively **high proportion** of saturated fatty acids
 - Easily stack, tending to be **solid** at room temperature
 - Contribute to **atherosclerosis** (动脉粥样硬化), in which lipid-containing **plaques** (斑块) build up along the inside walls of blood vessels (血管)



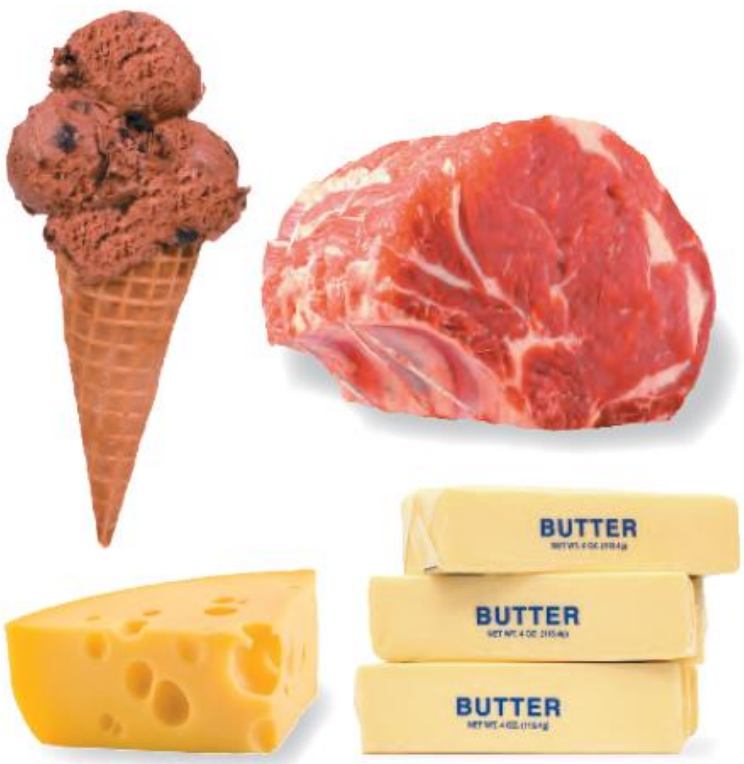
Unsaturated (不饱和的) Fats

- **Unsaturated**, if the carbon skeleton of a fatty acid has **fewer than the maximum** number of hydrogens at the double bond
- Monounsaturated
- Polyunsaturated
- Most plant and fish fats
 - high in unsaturated fatty acids
 - **liquid** at room temperature



Are all Unsaturated Fats Good for Health?

▼ **Figure 3.13** Types of fats.

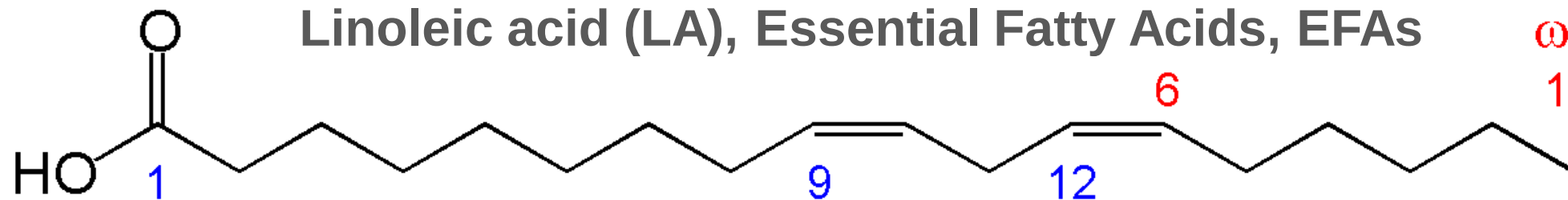
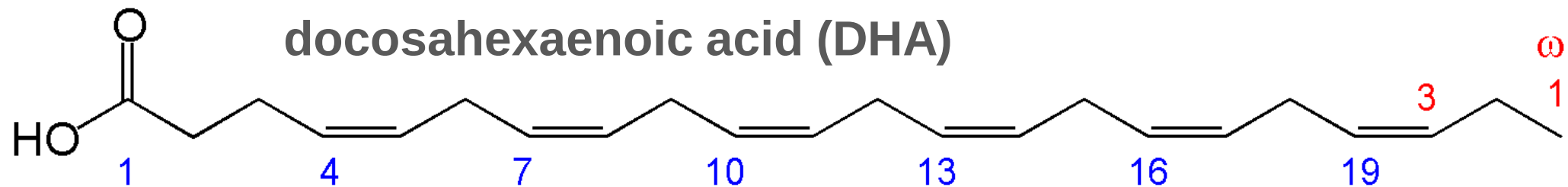
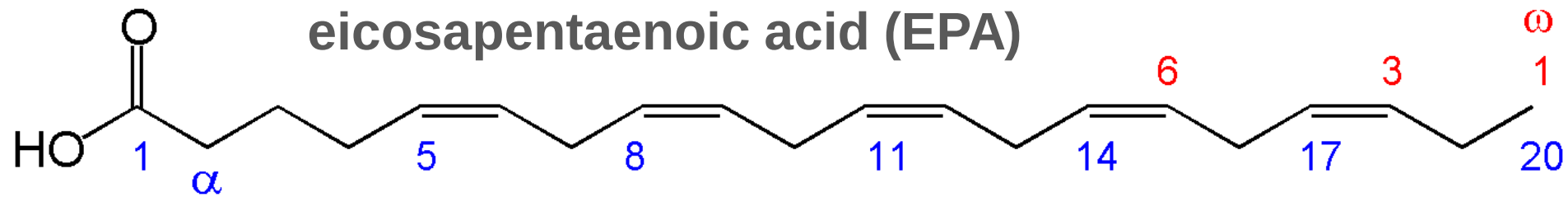
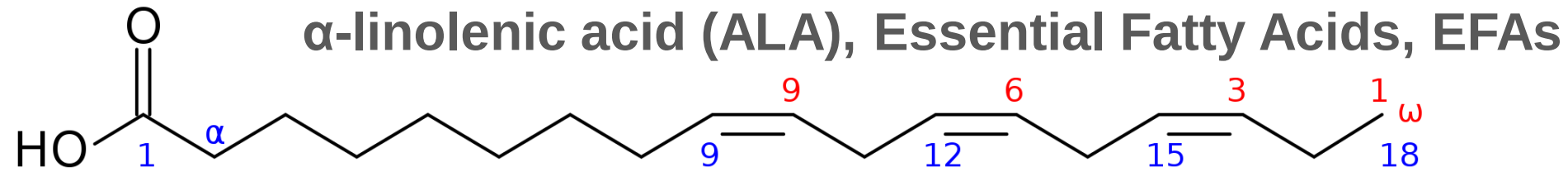
Saturated Fats (unhealthy fats found primarily in meat and full-fat dairy products; solid at room temperature)	Unsaturated Fats (fats found primarily in fish and plants; usually liquid at room temperature)
	 Plant and fish oils (unhydrogenated; usually liquid at room temperature)  INGREDIENTS: SOYBEAN OIL, FULLY HYDROGENATED COTTONSEED OIL, PARTIALLY HYDROGENATED COTTONSEED AND SOYBEAN OILS, MONO AND DIGLYCERIDES, TBHQ AND CITRIC ACID (ANTIOXIDANTS). Trans fats (unhealthy fats in hydrogenated processed foods; solid at room temperature)    Omega-3 fats (beneficial fats in some fish and plant oils; liquid at room temperature)

(反式脂肪)

Hydrogenation and Trans Fats

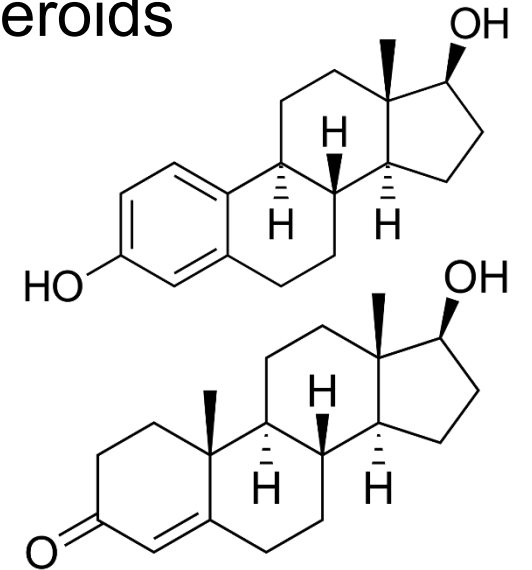
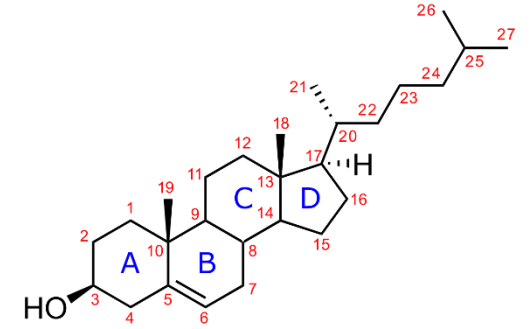
- “adds hydrogen”
- Converts unsaturated fats to saturated fats
 - Economic considerations
 - Stabilize the product
 - Extend the shelf life (保质期)
- Makes liquid fats solid at room temperature
- Creates **trans fatty acids (TFAs)**, a type of **unsaturated** fat that is **particularly bad** for your cardiovascular health

Omega-3 and Omega-6 fatty acids

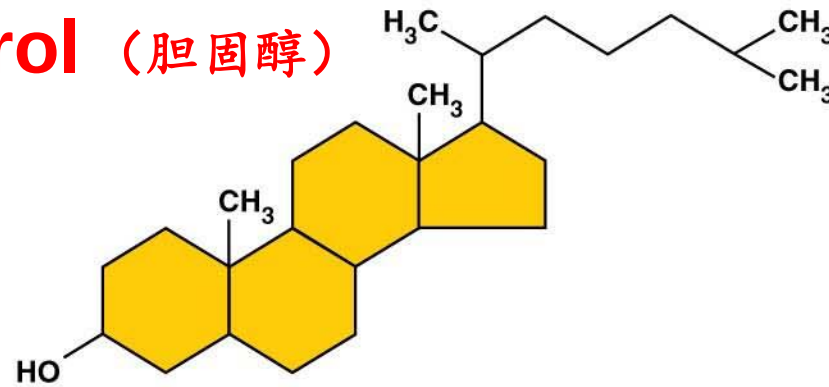


Steroids (类固醇)

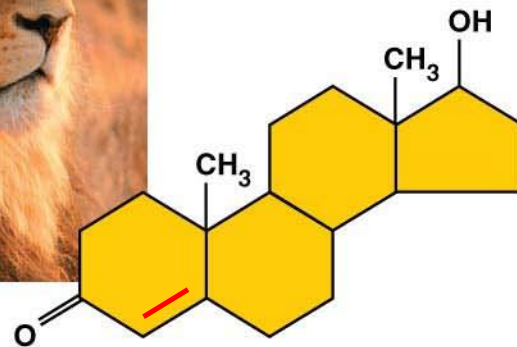
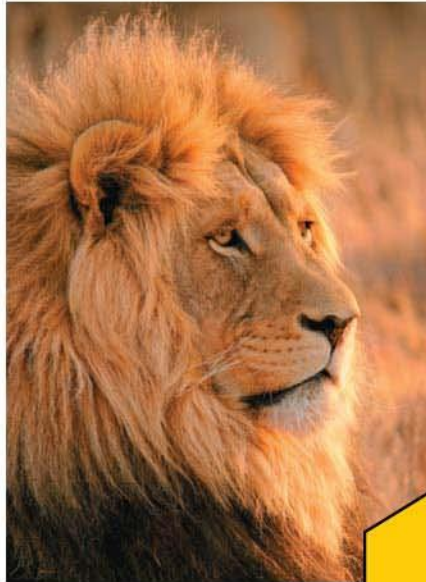
- Different from fats in structure and function
- Named after **Cholesterol** (胆固醇)
 - the “**base steroid**” from which your body produces other steroids
- Signaling molecules
 - Estradiol (雌二醇), an estrogen (雌激素)
 - Testosterone (睾酮)
- Important components of **cell membranes** that alter **membrane fluidity**



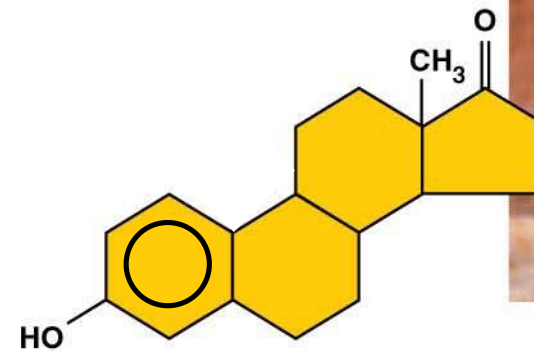
Cholesterol (胆固醇)



can be converted
by the body to



Testosterone



Estrone
(A type of estrogen)

Synthetic anabolic steroids (合成类固醇)

- Variants of testosterone, mimic some of its effects
- Originally prescribed to treat diseases such as cancer and AIDS
- Abused by athletes to build up their muscles quickly
- Serious physical and mental problems
- **Most** athletic organizations ban the use of anabolic steroids (stimulant) because of their many potential health hazards coupled with the unfairness of an artificial advantage.



Phospholipids (磷脂)

- A **hydrophilic** “head” with a negatively charged phosphate group
- Two nonpolar **hydrophobic** fatty acid “tails”.
 - One of the “tails” contains **cis bond**
 - EPA and DHA in marine phospholipids

